Moment-Weighted Fourier Envelopes for Heat Actions: A Reproducible Algorithm-Discovery Study

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Abstract

We investigate a proposed spectral-moment stopping rule for Chebyshev approximations to sparse symmetric diffusion actions. Positive Fourier polynomial identities yield low-degree square majorants of omitted coefficient blocks; standard dynamic programming selects a tail partition. Independent review traces the algebra and doubled-moment acquisition to established kernel polynomial methods, leaving application novelty unresolved. On six held-out case/tolerance settings, the rule saves 0–8.71% of matrix-vector products relative to ordinary Chebyshev. Its median end-to-end speed ratio against the fastest measured applicable baseline is 0.995, failing frozen 25% product and 1.5× speed targets. All 385 final validation/holdout timed outputs satisfy their empirical reference checks. An independently found floating-point counterexample narrows the supported numerical range; no full floating-point certificate is provided. Source snapshots, raw results and a fresh-environment reconstruction support reproduction of the negative result. A separately frozen continuation adds a linear-cost fixed-envelope evaluator and a faster exact dyadic checker. Its pipeline improves 4.46–8.01 times over unchanged v1 on the new cases, primarily through checker engineering; numerical-baseline comparisons remain mixed. Another 270 timed outputs pass their references.

1 Introduction

Applying a matrix function to a vector is a central operation in diffusion, network dynamics and exponential integration. For sparse symmetric positive semidefinite A, the action $\exp(-tA)b$ can be approximated without forming an exponential or storing an extensive basis. The question examined here is narrow: can already-computed information about the spectral measure of b reduce the work needed to justify truncating a Chebyshev expansion?

We study a moment-weighted stopping rule for Chebyshev heat actions. The construction uses established Fourier and kernel-polynomial identities and is valid in exact arithmetic. The experiments test whether sharper tail bounds yield practical savings after moment reductions and setup costs are included.

The evaluation combines an explicit candidate representation, independent reference computations, mathematical witnesses, matched-budget experiments and prospective comparisons with established numerical methods. Failed candidates and negative results are retained.

2 Related work and research positioning

AlphaTensor searches tensor factorizations and separates exact algebraic checking from hardware benchmarking [1]. AlphaDev explores bounded instruction programs [2]; its released agent description contains pseudocode rather than a complete execution system. FunSearch uses program

proposals and evaluator selection [3], while AlphaEvolve extends evolutionary coding to broader algorithmic tasks [4]. Computational Language Processing searches computational tokens using a grammar and reinforcement learning [5]. ERA combines code rewriting, tree search and external research ideas for empirical software [6]. None of their reported discoveries is a local replication in this report. We use no learned proposal distribution, paid model API or distributed training system.

For exponential actions, scaled truncated Taylor is a strong conventional method [7]. The experiments use SciPy's maintained public `expm_multiply`, rather than a simplified reconstruction. Restarted Krylov methods [8] are represented by SciPy's Hermitian implementation. A rational comparison uses the established CRAM16 incomplete partial factorization, with coefficients from OpenMC [9]. Its sparse factorizations and solves are included in one-shot cost.

Chebyshev exponential actions, memory advantages, spectrum sensitivity and error estimation have long been studied [10]. Residual-based Chebyshev stopping is established [11]; computable Krylov defect bounds are also established [12]. A May 2026 Chebyshev preprint constructs a short-recurrence residual estimator and adaptive enclosing ellipse [13]. Spectrum-adapted polynomial fitting predates this work [14]. Leja interpolation and adaptive rational Krylov are relevant additional methods; they were investigated but not benchmarked here. The comparisons therefore do not establish state-of-the-art superiority.

Most decisively, the kernel polynomial method (KPM) review [15], equations (34)–(35), contains the same doubled-moment identities used below; equations (56)–(58) contain the same coefficient-autocorrelation construction. Calling this a new "Pell majorant" would be misleading. The remaining application hypothesis is moment-weighted tail stopping with optimized contiguous partitions. Targeted independent literature searches did not locate an exact combined precedent, but absence of a match is not originality evidence.

3 Mathematical contract

The intended operation is

$$y = \exp(-tA)b, \quad A = A^T \succeq 0, \quad t \geq 0, \qquad \|\widehat{y} - y\|_2 \leq \varepsilon \|b\|_2. \tag{1}$$

The target is relative to input norm, not a possibly vanishing output norm. The implementation uses binary64 and $10^{-12} \leq \varepsilon \leq 10^{-3}$. Its checked API supports the sufficient domain of explicit, real, symmetric diagonally dominant matrices with nonnegative diagonal. Exact rational row inequalities on the stored binary64 coefficients establish that premise. Non-SDD PSD matrices refuse as unsupported, rather than being declared indefinite. Input conversion/canonicalization is part of the documented stored-matrix contract.

Let L be an outward-rounded Gershgorin upper bound, obtained from row sums. No oracle eigenvalues or Ritz lower bound are used as an upper enclosure. The $L = 0$, $t = 0$, and $b = 0$ cases are identities. For $L > 0$, set

$$B = 2A/L - I, \qquad z = tL/2, \qquad \sigma(B) \subset [-1, 1]. \tag{2}$$

The known Bessel generating function [16] gives

$$\exp(-tA)b = \sum_{k=0}^{\infty} c_k T_k(B) b, \qquad c_0 = e^{-z} I_0(z), \quad c_k = 2(-1)^k e^{-z} I_k(z) \ (k \geq 1). \tag{3}$$

The scaled special function $e^{-z} I_k(z)$ avoids exponential overflow.

Numerical support is narrower than the exact-arithmetic domain. An adversarial integer Laplacian kernel example in Section 8 violated the requested tolerance at large z. Before final evaluation the API therefore added the practical refusal policy

$$z \le \min\{20000, \varepsilon/(16ur)\}, \qquad \varepsilon\|b\|_2 \ge 64\,\mathrm{tiny}, \tag{4}$$

where u is binary64 machine epsilon, r is maximum CSR row width and tiny is the smallest positive normal binary64 value. These policies are implementation limits, not a proved forward-error theorem. They do not claim that inputs outside this range are mathematically ill posed. The returned bound controls ideal truncation under spectral premises; sampled binary64 accuracy and full floating-point certification remain distinct.

4 A moment-weighted Fourier-envelope stopping rule

Normalize b to input norm one for moment acquisition. For a consecutive omitted block beginning at $\ell \ge 1$, define

$$R(x) = \sum_{j=0}^{q-1} a_j T_{\ell+j}(x), \quad h_d = \sum_{j=0}^{q-1-d} a_j a_{j+d}, \quad Q(x) = h_0 + 2\sum_{d=1}^{q-1} h_d T_d(x). \tag{5}$$

This uses established positive Fourier-polynomial algebra [15].

Exact-arithmetic majorant. Put $x = \cos\theta$ and $F(\theta) = e^{i\ell\theta}\sum_j a_j e^{ij\theta}$. Then $R(x) = \Re F(\theta)$ and $Q(x) = |F(\theta)|^2$ by expanding the square and pairing conjugate terms. Consequently

$$Q(x) - R(x)^2 = (\Im F(\theta))^2 = (1-x^2)\left(\sum_{j=0}^{q-1} a_j U_{\ell+j-1}(x)\right)^2 \ge 0 \tag{6}$$

on $[-1,1]$, including endpoints by continuity. This proof is an identity, not a sampled positivity test. Exact-rational coefficient tests independently check it on several nontrivial blocks; they are supporting implementation checks, not a substitute for the derivation.

For $s_d = b^T T_d(B) b$, the spectral theorem integrates this pointwise bound:

$$\|R(B)b\|_2 \le \left(h_0 s_0 + 2\sum_{d=1}^{q-1} h_d s_d\right)^{1/2}. \tag{7}$$

Only moments through $q-1$ are needed, regardless of the block's carrier frequency ℓ. With $v_k = T_k(B)b$, established KPM identities give

$$s_{2k} = 2v_k^T v_k - s_0, \qquad s_{2k-1} = 2v_k^T v_{k-1} - s_1. \tag{8}$$

Thus a budget of p moments is acquired in roughly $p/2$ ordinary action iterations using reductions and scalar storage. This acquisition is not free: its runtime and memory are included in each one-shot candidate call.

Whole-tail accounting. Coefficients are computed through K. From the Bessel generating function, the symmetric Skellam moment-generating function is $\exp[z(\cosh\theta - 1)]$. Chernoff's inequality with $\theta = \operatorname{asinh}(k/z)$ and $k = K + 1$ yields

$$2e^{-z} \sum_{j=K+1}^{\infty} I_j(z) \le \eta_K := \min\left\{1, 2\exp\left[-k\operatorname{asinh}(k/z) + \frac{k^2}{\operatorname{hypot}(z,k) + z}\right]\right\}. \tag{9}$$

The rationalized last term avoids subtracting nearby large numbers. This bound covers every uncomputed coefficient, rather than declaring a finite sampled cutoff exact. Evaluation of special functions and this expression is still floating point, not interval arithmetic.

Partition synthesis. For each valid block $[i, i+q)$, $q \le p$, evaluate $w(i,q) = \sqrt{Q_{i,q}[s]}$. Starting with $C_{K+1} = \eta_K$, use

$$C_i = \min_{1 \le q \le \min(p, K+1-i)} \{w(i,q) + C_{i+q}\}. \tag{10}$$

This is standard shortest-path dynamic programming over contiguous partitions. The triangle inequality proves C_{m+1} bounds the omitted vector tail. Width-one blocks recover the coefficient-sum bound, which is also explicitly capped in execution. The reconstruction is a readable bound program with nonuniform block widths; it is not arbitrary generated source code.

The action recurrence remains the known Chebyshev recurrence. The proposed change is the mathematical stopping rule. Its leaves and optimization are established ingredients; the engine did not independently invent their underlying identities. The primary researcher and specialists supplied those hypotheses, and the engine instantiated and evaluated their compositions.

Cost and numerical caveats. For K coefficients and p moments, vectorized block formation costs $O(Kp^2)$ scalar work, followed by $O(Kp)$ dynamic programming, and uses $O(Kp)$ scalar workspace plus $O(n)$ live vectors. Every padded block actually evaluated is counted, even when not admissible in the DP. This setup can exceed the cost of the few sparse products it saves. Cancellation in $Q[s]$ receives a practical summation margin; materially negative computed values refuse that block. This margin does not enclose propagated moment, sparse-product, coefficient or recurrence rounding error. No floating-point certificate follows.

5 Discovery experiments and independent verification

A typed data representation permits finite Fourier-envelope blocks, coefficient sum remainders, an optional two-anchor transport bound, addition through the triangle inequality, and minima of valid bounds. It validates initialized integer widths, moment requirements, schema fields and finite work budgets. Width-one equivalents are recognized. No proposed Python is evaluated, no candidate controls filesystem/network operations, and no unbounded candidate loops are admitted. This is data interpretation, not a hostile-code OS sandbox.

An exact calibration enumerates 343 coefficient triples in $v_{k+1} = (ax + c)v_k + bv_{k-1}$ under fixed degree-two/three constraints. The sole survivor is $(a, b, c) = (2, -1, 0)$, the known Chebyshev recurrence. This is deliberately reported as rediscovery. It does not demonstrate unconstrained mathematical creativity or a new theorem prover.

Four bounded proposal policies receive 24 attempts per seed and three fixed seeds: enumeration, random proposals, evolutionary mutations and a bottleneck-guided mutation. The historical last identifier is `counterexample`; no accuracy counterexample occurred in that search, so the label

does not establish counterexample-guided synthesis. Duplicates consume proposal budget. All policies reach the same development product-count score, 370, with 265 verified proposals and 23 duplicates across 288 attempts. The deterministic policy's repeated seeds are repeated costs, not independent stochastic search outcomes. No learned proposal policy is added to conceal this tie. DP subproblems and full candidate evaluations have different units; we do not call one DP solve one candidate in a superiority comparison.

Independent reference code uses 80-digit `mpmath.eigsy` on small matrices, SciPy dense matrix exponentials as a different computational path, and analytic Dirichlet eigenvalues with DST-I on large separable grids. Semigroup, contractivity, repeated/zero modes, signed SDD matrices, extreme scales and explicit rejection are checked. Large nonseparable graphs use a tight Taylor/CRAM crosscheck, which is empirical consistency, not certified truth. Accuracy of those two baseline methods on that crosscheck is not independently established by agreement with their own reference.

6 Experimental protocol

Development uses one-dimensional diffusion and diagonal spectral clusters; validation uses two-dimensional diffusion and weighted cycle/chord graphs. The final evaluation uses anisotropic diffusion, disconnected heterogeneous diffusion blocks and signed SDD operators. Final families are excluded from search and its scores. Two tolerances on the same matrix/vector are paired settings, not independent dataset replications. Seeds, dimensions, CSR/input checksums, source snapshots, library versions and raw errors/timings are saved.

The criterion was frozen after initial development calibration and before final evaluation: zero tolerance misses, at least 25% fewer matrix-vector products and a 1.5× median end-to-end improvement over the strongest applicable baseline on two held-out families, with no more than 10% regressions elsewhere for a broad claim. A result failing those targets remains a failure; the thresholds are not retuned after looking at holdout.

Each method is warmed, then run seven times in randomized order. Coefficient generation, moment estimation, DP setup, sparse factorizations, selection and allocations remain inside the measured solve. Exact domain validation is common to all methods and recorded separately, with totals including it. Each timed output is independently checked after stopping its timer; worst error is retained. Operation counts come from separate diagnostic runs. A CSR subclass preserves SciPy's sparse norm-selection path; an earlier LinearOperator count path was rejected in review. Public Taylor/CRAM defaults target substantially higher accuracy than requested-epsilon methods, a stated practical API comparison rather than a matched-tolerance optimality claim.

The hardware is the available arm64 macOS machine, ten reported logical CPUs. A hardware inventory query was denied; total RAM and CPU model were not measured and no alternate route bypassed that denial. BLAS thread limits are requested through `VECLIB_MAXIMUM_THREADS`, `OPENBLAS_NUM_THREADS`, `OMP_NUM_THREADS` and threadpoolctl. Accelerate's actual thread count is not independently observed. The machine is not reserved; seven samples and within-case bootstrap intervals do not establish hardware-isolated statistics. Exploratory timings overlapping search are preserved but excluded from final runtime conclusions. No GPU measurements are claimed.

SuiteSparse's official collection reported downloads offline. All benchmark matrices here are documented generated structural families, not fabricated real-data coverage. The largest separable grid and all graph sizes are stated in the generated result tables. One-shot performance is primary; no setup amortization advantage is asserted without separate measurements.

Table 1: Median solve latency in milliseconds, including method setup. Common exact-domain validation is reported separately in raw results. Each row has seven randomized-order repetitions per method.

Case	n	ε	Cheb.	Moment DP	Taylor	Krylov	CRAM16
Aniso.	9216	10^{-6}	12.32	14.37	187.07	28.82	518.53
Aniso.	9216	10^{-10}	22.97	37.18	330.36	62.76	931.68
Aniso. stiff	16384	10^{-6}	83.62	80.27	3470.65	243.05	1742.07
Aniso. stiff	16384	10^{-10}	136.84	107.13	3267.03	514.02	2021.12
Signed SDD	4096	10^{-10}	4.45	10.68	74.10	19.10	13248.34
Disjoint	4096	10^{-10}	80.68	66.85	1997.61	416.56	68.63

7 Results

The final evaluation contains 6 paired case/tolerance settings and 30 method settings, with 210 timed outputs. All final validation and holdout outputs pass their empirical reference checks. The largest observed candidate error is 0.604 of its tolerance. Product savings relative to ordinary Chebyshev range from 0.00% to 8.71%. The median end-to-end speed ratio against the fastest measured applicable baseline is 0.995; a ratio below one means the candidate is slower. The frozen 25% product-reduction and 1.5× speed targets fail. No original-discovery success is declared.

The recorded proposal experiment consumed 55.08 seconds observed wall time and 98,365 matrix-vector products across all attempted evaluations. No learned proposal model was trained and no paid model calls occurred. Its overlap with exploratory validation prevents interpreting that wall time as a controlled comparison among search policies. An offline validation ablation using identical low moments compares optimal contiguous DP, fixed-width and greedy partitions. DP saves at most 0 additional degrees relative to fixed-width in these 14 settings; full rows are retained. All methods in this ablation share an artificial minimum-degree start, so those numbers are not substituted for production timings. Fresh-process peak RSS, including interpreter, libraries, input and solve, is chebyshev: 59.7 MiB, moment-dp: 60.6 MiB, scipy-taylor: 60.0 MiB, scipy-krylov: 68.0 MiB, cram16: 82.0 MiB. These absolute process peaks are not incremental algorithm allocation measurements.

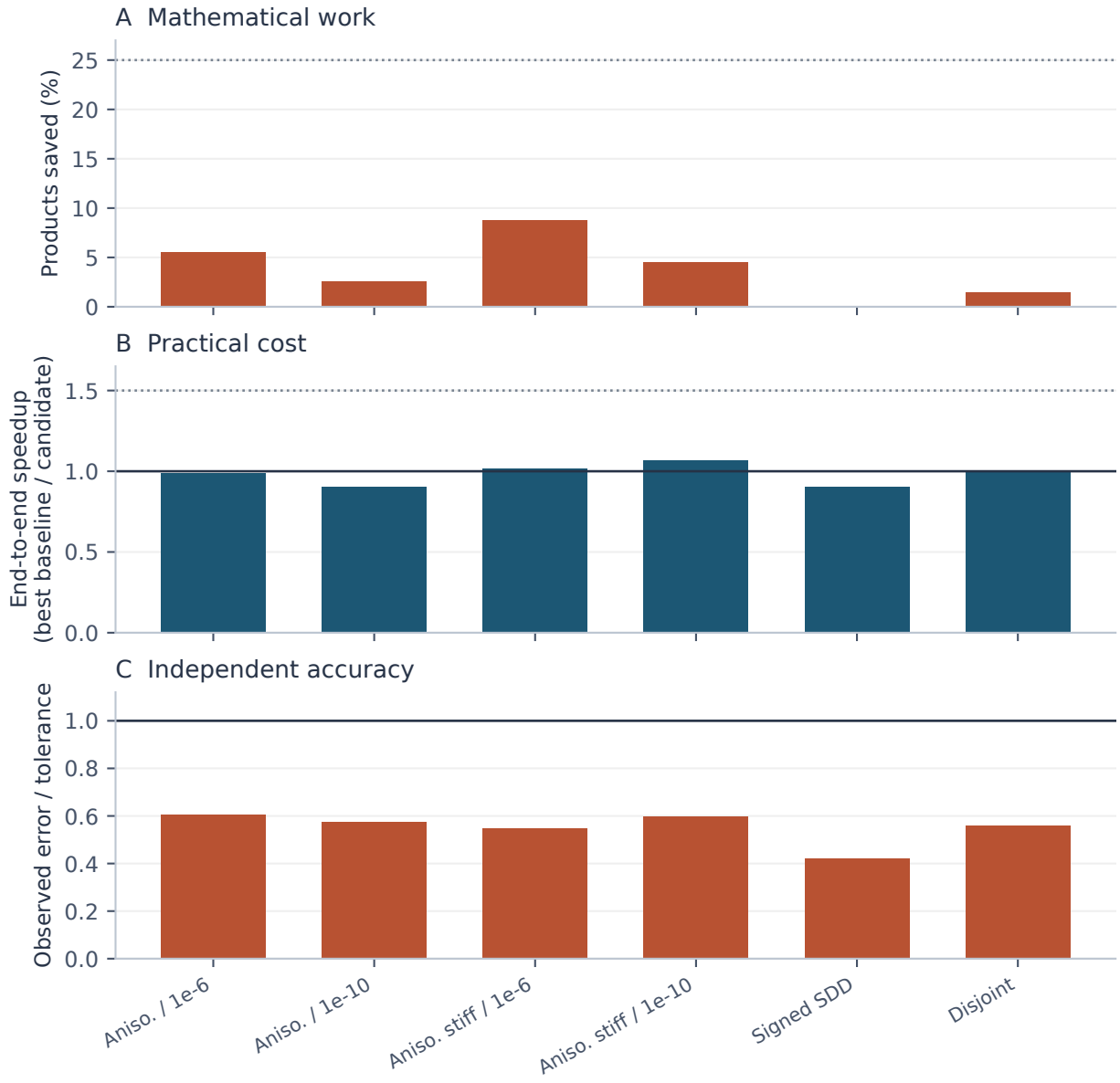


Figure 1: Actual held-out evidence. Dashed lines mark the frozen stretch thresholds; one is the accuracy acceptance limit. Costs include common domain validation in panel B. Signed-SDD accuracy uses a Taylor/CRAM crosscheck, not certified truth.

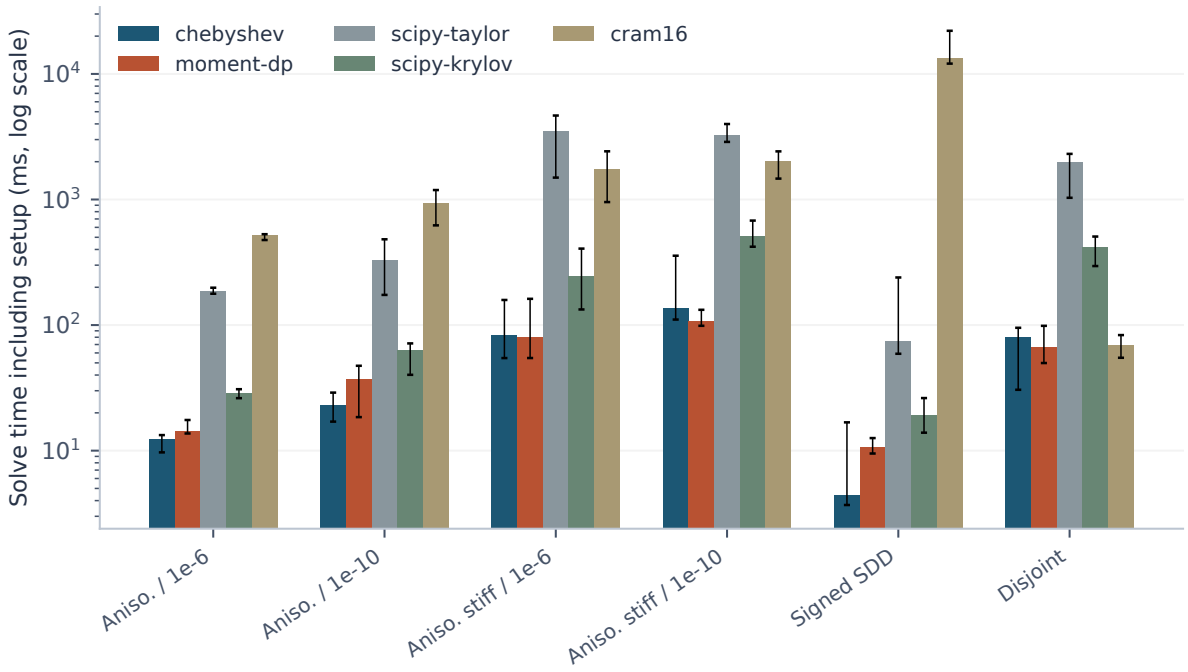


Figure 2: Solve latency with within-case bootstrap median intervals. Public Taylor/CRAM accuracy policies are stricter than requested epsilon. CRAM includes eight complex factorizations, explaining its unfavorable one-shot graph cost. These are CPU measurements on one unreserved machine, not cross-hardware conclusions.

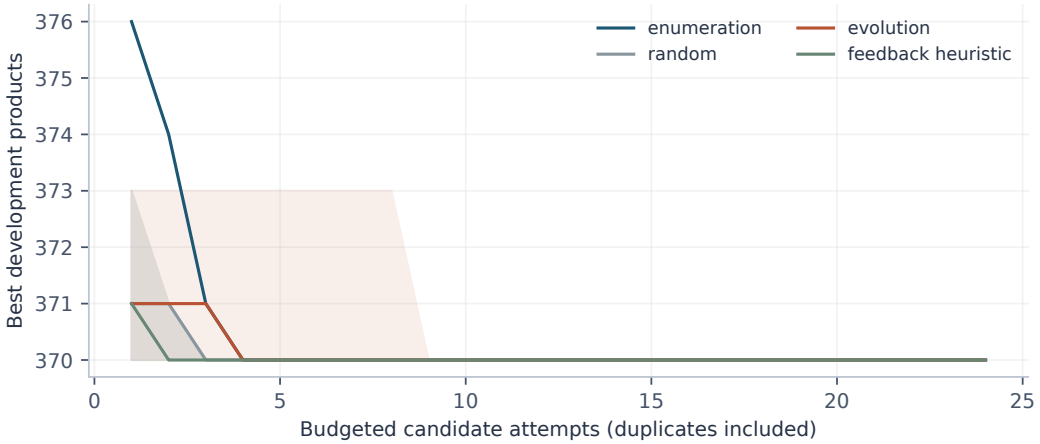


Figure 3: Recorded equal-proposal-budget search. Every policy reaches 370 development products; bands show the three saved seeds. The feedback identifier did not encounter accuracy counterexamples. No learned or evolutionary search superiority is supported.

8 Adversarial failures and limitations

The first two-anchor recurrence bound failed to improve broad spectral measures. It was retained as a useful negative hypothesis rather than promoted by a carefully chosen singleton example. The moment-envelope application reduces some product counts, but this is a change in stopping control, not a newly invented exponential expansion.

A genuine numerical counterexample was found independently:

$$A = \begin{pmatrix} 115061 & -54638 & -60423 \\ -54638 & 108974 & -54336 \\ -60423 & -54336 & 114759 \end{pmatrix}, \quad b = (1,1,1)^T, \quad t = 39800/230122. \tag{11}$$

The exact stored matrix annihilates b, so the exact answer is b. The pre-amendment implementation at $\varepsilon = 10^{-12}$ returned after 1008 products, with normalized truncation bound $8.78193 \cdot 10^{-13}$ but actual normalized error $1.08609 \cdot 10^{-12}$. This violated the target under a valid PSD premise. The problem is accumulated floating-point error, not a false square-majorant identity. It triggered the explicit support refusal above; it is not hidden by widening tolerance. A subnormal target counterexample similarly triggered rejection rather than a spurious zero-bound answer.

The final implementation is not formally floating-point certified. Its practical support guard is not a forward-error theorem. A fully certified algorithm would need interval-enclosed coefficients, propagated moment and recurrence errors, and rounding-aware spectral transformations. General PSD matrices outside SDD, unbounded stiff scales, block right-hand sides, distributed execution and GPUs are not implemented. The benchmark has few matrices, generated families, limited stochastic replications, and no current residual-Chebyshev/Leja head-to-head comparison. These are concrete limits on a research claim, not completed work.

9 Continuation: removing costs without changing the claim

A separately frozen November 2 protocol tested two engineering changes on new 3D Dirichlet diffusion, Neumann diffusion with reaction, and signed heterogeneous ladder operators. The v1 source and all earlier evidence were preserved. The existing DP default remains available; fixed-width evaluation is explicit.

9.1 Sliding fixed-width envelopes

Let coefficients be zero-padded and write $Q_i = c_{i:i+p}^T \,\mathrm{Toeplitz}(s) c_{i:i+p}$. Direct expansion gives

$$\begin{aligned} Q_i = Q_{i+1} + s_0(c_i^2 - c_{i+p}^2) + 2c_i \sum_{d=1}^{p-1} s_d c_{i+d} \\ - 2c_{i+p} \sum_{d=1}^{p-1} s_d c_{i+p-d}. \end{aligned} \tag{12}$$

Correlations compute the boundary sums. Reverse accumulation from the padded end obtains every block form in $O(Kp)$ scalar arithmetic and $O(K+p)$ space; residue-class sums give the fixed-partition tail. This replaces the optional DP's $O(Kp^2)$ construction. The practical scalar margin accumulates absolute boundary terms in the same direction. It still excludes errors in moment acquisition, special functions and sparse recurrence execution.

Table 2: Continuation median end-to-end latency (ms), including exact premise checking. Unchanged v1 is an engineering control, not the best conventional numerical baseline.

Case	v1 pipeline	Chebyshev	DP/new check	Fixed/new check	Krylov
3D Dirichlet / 1e-6	333.72	53.85	57.86	47.40	18.50
3D Dirichlet / 1e-10	260.20	54.22	48.59	43.62	60.11
Neumann + reaction / 1e-6	227.77	31.75	34.40	28.43	63.87
Neumann + reaction / 1e-10	237.09	38.18	45.99	42.48	74.55
Signed ladder / 1e-6	44.62	7.00	8.80	9.58	15.09
Signed ladder / 1e-10	50.00	7.05	13.70	11.22	19.15

Fixed width is one admissible partition, not mathematically equivalent to DP. For example, a valid singleton spectral measure at $x = 1/2$, $z = 1000$, width 8 and tolerance 10^{-10} admits DP stopping one degree earlier. Other singleton examples have larger gaps. Such examples do not establish a practical DP win against Lanczos, which resolves a singleton immediately. Incremental quadratic forms and this fixed partition are engineering mechanisms, not presumed new mathematical discoveries.

9.2 A faster exact premise check

When scaling by a tested power of two turns every stored matrix coefficient into an integer, and maximum integer magnitude times maximum row width is at most 2^{53}, every positive row-sum intermediate is exactly representable in binary64. Summation order cannot change that integer result. Diagonal dominance and its spectral row-sum bound are therefore checked exactly on that lattice. Inputs outside this sufficient fast path retain general Fraction arithmetic. Empty rows, integer boundaries, signed edges and false dominance are tested against an independent rational implementation. This optimization preserves the SDD premise; it does not turn a truncation bound into a floating-point certificate.

9.3 Prospective evidence

The new run uses independent sine/cosine spectral transforms on the grids and a symmetric band eigendecomposition on the ladder. Tiny explicit adjacency matrices were checked against 80-digit eigensolves. Every timed result is validated after its timer. The v1 engineering control imports unchanged source-snapshotted code on identical inputs; other methods receive the same new premise checker. Nine randomized-order paired repetitions are retained. The 30 method settings produce 270 accurate timed outputs. Fixed mode uses the same stopping degree as DP in all six new settings, but that empirical tie is not a theorem. Its complete pipeline is 4.46–8.01 times faster than the unchanged v1 pipeline, largely because the old rational premise check dominated cost. The maximum observed fixed-mode error is 0.631 of its tolerance.

Against the best measured ordinary-Chebyshev or maintained-Krylov baseline per case, the median total-time ratio is 0.815. Results remain mixed, and product savings reach only 9.09%. The original scientific thresholds still fail. The engineering-control speedups must not be presented as a new numerical algorithm beating established mathematics. Taylor, CRAM16, Leja and current residual-Chebyshev are not newly benchmarked on these families. A fresh dependency environment reconstructs the new source-snapshotted action; its matrix/vector checksum and product count match. The expanded independent suite contains 88 passing checks. No core identity, exactness premise or originality claim is strengthened merely because these engineering changes work.

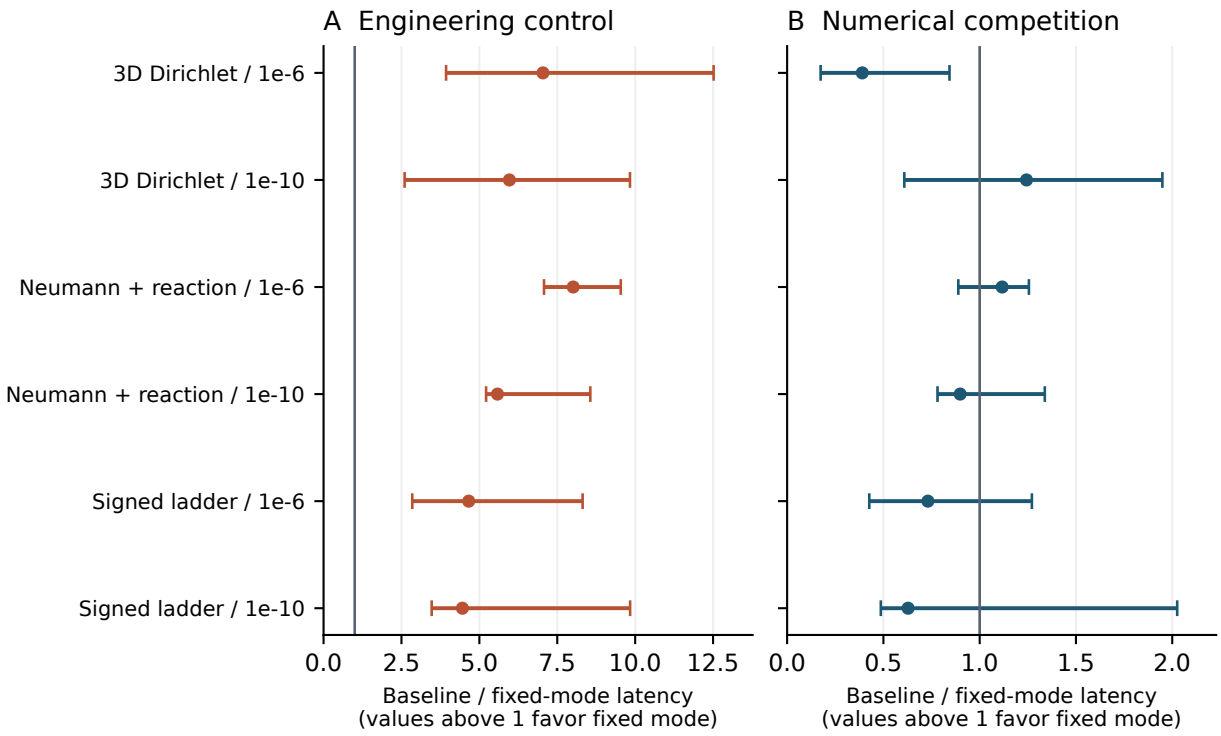


Figure 4: New measured latency ratios, with paired bootstrap intervals. Panel A compares unchanged v1 engineering cost. Panel B compares the best measured Chebyshev/Krylov baseline under the same new exact checker. The latter does not show consistent superiority. Intervals describe nine repetitions on one unreserved machine, not hardware-isolated certainty.

10 Discussion and conclusion

Low-order moments can sharpen a coefficient-sum stopping bound without additional action iterations. In the evaluated settings, reductions and scalar setup can offset product savings, particularly for broad-spectrum operators. Product count alone is therefore insufficient to predict end-to-end cost.

The central identities and moment acquisition were independently traced to prior KPM work. The DP is standard optimization. The narrower tail-stopping application remains a research hypothesis assembled from those ingredients, whose practical advantage remains unestablished. Further work requires a stable mechanism whose advantage survives strong adaptive baselines, unseen workloads and independent reproduction.

Reproduction and provenance. The repository supplies exact dependencies, CLI/API, raw ledgers, source snapshots, figure scripts, mathematical witnesses, refusal tests and reviewer findings. The retained records support regeneration of the reported figures and numerical comparisons.

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